Problem (`baseline_calcs`): Compute the variance of the policy gradient estimator (5 points)

Let π_θ define a policy over the binary action space $\mathcal{A} = \{0, 1\}$ with $\pi_\theta(A = 1) = p = \sigma(\theta)$, where σ represents the sigmoid function $\sigma(\theta) = \frac{1}{1+e^{-\theta}}$. Our binary reward function will assign reward 1 to action $A = 1$ and reward 0 otherwise, or $r(A) = \mathbf{1}\{A = 1\}$.

(a) Let the policy gradient estimator be given by

$$\frac{1}{n}\sum_{i=1}^{n} r(A_i)\,\nabla_\theta \log \pi_\theta(A_i) \tag{18}$$

given n samples $A_i \overset{\text{iid}}{\sim} \pi_\theta$. What is the variance of this estimator?

Deliverable: An expression in terms of n and p, accompanied by a derivation.

$$\mathrm{Var}(\hat{g}) = \frac{p(1-p)^3}{n}.$$

(b) Let the baseline-adjusted policy gradient estimator be given by

$$\frac{1}{n}\sum_{i=1}^{n} \big(r(A_i) - b\big)\,\nabla_\theta \log \pi_\theta(A_i) \tag{19}$$

given n samples $A_i \overset{\text{iid}}{\sim} \pi_\theta$. What is the variance of this estimator?

Deliverable: An expression in terms of n, b, and p, accompanied by a derivation and some discussion.

$$\mathrm{Var}(\hat{g}_b) = \frac{p(1-p)}{n}\big[b - (1-p)\big]^2.$$

(c) What is the resulting variance if we substitute the "population mean" baseline $b = p$? Compare this variance to that of the unadjusted policy gradient estimator: is it always lower, always higher, or sometimes higher or lower depending on p?

$$\mathrm{Var}(\hat{g}_p) = \frac{p(1-p)}{n}(1-2p)^2,$$

$$\mathrm{Var}(\hat{g}_p) - \mathrm{Var}(\hat{g}) = \frac{p^2(1-p)}{n}(3p-2).$$

$$\mathrm{Var}(\hat{g}_p) \begin{cases} < \mathrm{Var}(\hat{g}), & 0 < p < \frac{2}{3}, \\ = \mathrm{Var}(\hat{g}), & p = \frac{2}{3}, \\ > \mathrm{Var}(\hat{g}), & \frac{2}{3} < p < 1. \end{cases}$$